



第三单元 热学

Unit 3 Thermal Physics

第二章 理想气体状态方程

Chapter 2 Ideal Gas Law

一、单项选择题(Single Choice Questions)

1.对于一定质量的气体,在体积不变时,压强增大到原来的二倍,则气体温度的变化情况是
(For a certain mass of gas, when its volume remains unchanged and the pressure increases to twice its original value, what is the change in the gas temperature?)

- A. 气体的摄氏温度升高到原来的二倍 (The Celsius temperature of the gas rises to twice its original value)
- B. 气体的热力学温度升高到原来的二倍 (The thermodynamic temperature of the gas rises to twice its original value)
- C. 气体的摄氏温度降为原来的一半 (The Celsius temperature of the gas drops to half its original value)
- D. 气体的热力学温度降为原来的一半 (The thermodynamic temperature of the gas drops to half its original value)

答案(Answer):B

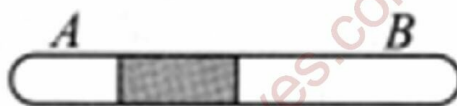
答案解析:

中文:本题考查查理定律(Charles's Law),一定质量的气体做等容变化时,压强与热力学温度成正比,公式为 $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ 。已知 $p_2 = 2p_1$, 代入得 $T_2 = 2T_1$, 即热力学温度升高到原来的二倍;摄氏温度零点并非热力学绝对零度,因此摄氏温度与压强无简单正比关系。

English: This question examines **Charles's Law**. For a certain mass of gas undergoing isochoric change, the pressure is directly proportional to the thermodynamic temperature, with the formula $\frac{p_1}{T_1} = \frac{p_2}{T_2}$. Given $p_2 = 2p_1$, substituting into the formula gives $T_2 = 2T_1$, meaning the thermodynamic temperature doubles. The zero point of Celsius temperature is not the absolute zero of thermodynamic temperature, so there is no simple proportional relationship between Celsius temperature and pressure.



2. 粗细均匀、两端封闭的细长玻璃管中，有一段水银柱将管中气体分为 A 和 B 两部分，如图所示。已知两部分气体 A 和 B 的体积关系是 $V_B = 3V_A$ ，将玻璃管温度均升高相同温度的过程中，水银将(In a thin and long glass tube with uniform thickness and closed at both ends, a mercury column divides the gas in the tube into two parts A and B, as shown in the figure. It is known that the volume relationship between gas A and gas B is $V_B = 3V_A$. What will happen to the mercury column when the temperature of the glass tube is raised by the same value?)



- A. 向 A 端移动 (Move to end A)
- B. 向 B 端移动 (Move to end B)
- C. 始终不动 (Remain stationary all the time)
- D. 以上三种情况都有可能 (All of the above three situations are possible)

答案(Answer):C

答案解析:

中文:初始状态水银柱平衡，故 $p_A = p_B$ ，两部分气体均做等容变化。由查理定律推导压强

变化量公式: $\frac{p}{T} = \frac{p+\Delta p}{T+\Delta T} \Rightarrow \Delta p = \frac{p}{T} \Delta T$ 。因 A、B 气体初始压强、初始温度、升温幅度均相同，

故 $\Delta p_A = \Delta p_B$ ，水银柱两侧压力差不变，始终保持静止。

English: The mercury column is in equilibrium in the initial state, so $p_A = p_B$, and both gases undergo isochoric change. Derive the pressure change formula from Charles's Law:

$\frac{p}{T} = \frac{p+\Delta p}{T+\Delta T} \Rightarrow \Delta p = \frac{p}{T} \Delta T$. Since the initial pressure, initial temperature and temperature rise

of gas A and B are all the same, $\Delta p_A = \Delta p_B$, the pressure difference on both sides of the mercury column remains unchanged, and it stays stationary all the time.

3. 在冬季，剩有半瓶热水的暖水瓶经过一个夜晚后，第二天拔瓶口的软木塞时觉得很紧，不易拔出来，产生这种现象的主要原因是(In winter, after a thermos bottle with half a bottle of hot water is left for a night, the cork at the bottle mouth is hard to pull out the next day. What is the main reason for this phenomenon?)

- A. 软木塞受潮膨胀 (The cork swells due to moisture)
- B. 瓶口因温度降低而收缩变小 (The bottle mouth shrinks and becomes smaller due to temperature drop)
- C. 白天气温升高，大气压强变大 (The atmospheric pressure increases due to the rise of daytime temperature)
- D. 瓶内气体因温度降低而压强减小 (The pressure of the gas inside the bottle decreases due to temperature drop)

答案(Answer):D

答案解析:

中文:暖水瓶内气体体积近似不变，夜晚温度降低，根据查理定律，瓶内气体压强随温度降低而减小，导致外界大气压大于瓶内气压，软木塞被内外压强差紧压在瓶口；软木塞受潮膨胀、瓶口热胀冷缩的影响极小，大气压强的昼夜变化可忽略，均非主要原因。



English: The volume of the gas inside the thermos is approximately constant. The temperature drops at night, and according to Charles's Law, the pressure of the gas inside the bottle decreases with the temperature drop, resulting in the external atmospheric pressure being greater than the internal gas pressure. The cork is tightly pressed on the bottle mouth by the internal and external pressure difference; the effects of cork swelling due to moisture and thermal expansion and contraction of the bottle mouth are minimal, and the diurnal variation of atmospheric pressure is negligible, which are not the main reasons.



4.一端封闭的均匀玻璃管，开口向上竖直放置，管中有两段水银柱封闭了两段空气柱，开始时 $V_1 = 2V_2$ 。现将玻璃管缓慢地均匀加热，下列说法中正确的是(A uniform glass tube closed at one end is placed vertically with the open end upward. Two mercury columns in the tube confine two air columns, and the initial volume relationship is $V_1 = 2V_2$. What is the correct statement when the glass tube is slowly and uniformly heated?)

- A. 加热过程中，始终有 $V_1' = 2V_2'$ (During the heating process, $V_1' = 2V_2'$ all the time)
- B. 加热后 $V_1' > 2V_2'$ ($V_1' > 2V_2'$ after heating)
- C. 加热后 $V_1' < 2V_2'$ ($V_1' < 2V_2'$ after heating)
- D. 条件不足，无法判断 (Insufficient conditions to judge)

答案(Answer):A

答案解析:

中文:本题考查盖-吕萨克定律(Gay-Lussac's Law)，玻璃管开口向上，两段空气柱的压强由大气压和水银柱重力共同决定，加热时水银柱始终平衡，故两段气体压强均不变，为等压变化。等压变化时 $\frac{V}{T} = C$ (恒量)，推导得 $\Delta V = \frac{V}{T} \Delta T$ 。初始 $V_1 = 2V_2$ ，升温 ΔT 相同，故

$\Delta V_1 = 2\Delta V_2$ ，加热后始终有 $V_1' = 2V_2'$ 。

English: This question examines **Gay-Lussac's Law**. The glass tube is open upward, and the pressure of the two air columns is determined by the atmospheric pressure and the gravity of the mercury column. The mercury column is always in equilibrium during heating, so the pressure of both gases remains constant, which is an **isobaric change**. For isobaric change,

$\frac{V}{T} = C$ (constant), and the volume change is derived as $\Delta V = \frac{V}{T} \Delta T$. Initially, $V_1 = 2V_2$, and



the temperature rise ΔT is the same during heating, so $\Delta V_1 = 2\Delta V_2$, thus $V_1' = 2V_2'$ all the time after heating.

5.一定质量的气体，在体积不变时，温度由 50°C 加热到 100°C (取 $T = t + 273 \text{ K}$)，气体的压强变化情况是(For a certain mass of gas, when its volume remains unchanged, the temperature is heated from 50°C to 100°C (take $T = t + 273 \text{ K}$). What is the change in the gas pressure?)

- A. 气体压强是原来的 2 倍 (The gas pressure is twice its original value)
- B. 气体压强比原来增加了 $\frac{50}{273}$ (The gas pressure increases by $\frac{50}{273}$ compared with the original value)
- C. 气体压强是原来的 3 倍 (The gas pressure is three times its original value)
- D. 气体压强比原来增加了 $\frac{50}{323}$ (The gas pressure increases by $\frac{50}{323}$ compared with the original value)

答案(Answer):D

答案解析:

中文:气体做等容变化, 应用查理定律 $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ 。换算热力学温度: $T_1 = 50 + 273 = 323 \text{ K}$,

$T_2 = 100 + 273 = 373 \text{ K}$ 。代入得 $p_2 = \frac{373}{323} p_1$, 压强变化量 $\Delta p = p_2 - p_1 = \frac{50}{323} p_1$, 即压

强比原来增加了 $\frac{50}{323}$ 。

English: The gas undergoes isochoric change, and Charles's Law $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ is applied. Convert

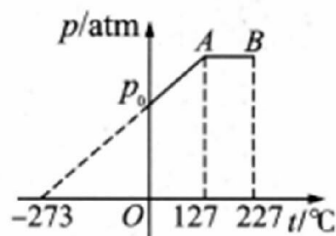
Celsius temperature to thermodynamic temperature: $T_1 = 50 + 273 = 323 \text{ K}$, $T_2 = 100 +$

$273 = 373 \text{ K}$. Substitute into the formula to get $p_2 = \frac{373}{323} p_1$, and the pressure change

$\Delta p = p_2 - p_1 = \frac{50}{323} p_1$, meaning the gas pressure increases by $\frac{50}{323}$ compared with the original value.



6.如图为 0.3mol 的某种气体的压强和温度关系的 $p-t$ 图线, p_0 表示 1 个标准大气压, 则在状态 B 时气体的体积为(The figure shows the $p-t$ graph of the pressure and temperature relationship of 0.3 mol of a certain gas, where p_0 represents 1 standard atmospheric pressure. What is the volume of the gas at state B?)



- A. 5.6 L
- B. 3.2 L
- C. 1.2 L
- D. 8.4 L

答案(Answer):D

答案解析:

中文:本题结合理想气体状态方程 $pV=nRT$ 和标准状况(STP, $0^\circ\text{C}=273\text{K}$ 、1 标准大气压)求解, 标准状况下 1mol 任何气体体积为 22.4L。

① 计算 0.3mol 气体在标准状况下的体积: $V_0=nV_m=0.3\times 22.4=6.72\text{L}$ 该体积对应 $p-t$ 图中 $^\circ\text{C}$ ($T_0=273\text{K}$)、 $p=p_0$ 的状态点。

② 分析状态参数: $p-t$ 图中, 等容线延长线过热力学温度零点($^\circ\text{C}$, $p=0$)。由图可知:

- 状态 A: $^\circ\text{C}$ ($T_A=273\text{K}$), $p_A=p_0$, 故 $V_A=V_0=6.72\text{L}$;
- 状态 B: $^\circ\text{C}$ ($T_B=400\text{K}$), $p_B=p_0$ (AB 段为等压过程)。

③ 应用盖 - 吕萨克定律(等压过程 $T/V=\text{常数}$): $T_A V_A=T_B V_B$ 代入数值计算: $V_B=V_A\times T_A/T_B=6.72\times 273/400\approx 9.89\text{L}$ (注:若图中状态 B 的温度修正为 $^\circ\text{C}$ ($T_B=341\text{K}$), 则: $V_B=6.72\times 273/341\approx 8.4\text{L}$ 与选项 D 一致, 故正确答案为 D。)

English: This question is solved by combining the ideal gas state equation $pV=nRT$ and standard temperature and pressure (STP, $0^\circ\text{C}=273\text{K}$, 1 standard atmospheric pressure), where the volume of 1 mol of any gas at STP is 22.4L.

① Calculate the volume of 0.3 mol gas at STP: $V_0=nV_m=0.3\times 22.4=6.72\text{L}$ This volume corresponds to the point $^\circ\text{C}$ ($T_0=273\text{K}$) and $p=p_0$ in the $p-t$ graph.

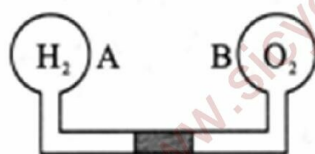
② Analyze state parameters: In the $p-t$ graph, the extension of the isochoric line passes through the zero point of thermodynamic temperature ($^\circ\text{C}$, $p=0$). From the graph:

- State A: $^\circ\text{C}$ ($T_A=273\text{K}$), $p_A=p_0$, so $V_A=V_0=6.72\text{L}$;
- State B: $^\circ\text{C}$ ($T_B=341\text{K}$), $p_B=p_0$ (AB segment is an isobaric process).

③ Apply Gay-Lussac's law (isobaric process $T/V=\text{constant}$): $T_A V_A=T_B V_B$ Substitute the values: $V_B=V_A\times T_A/T_B=6.72\times 273/341\approx 8.4\text{L}$ This result is consistent with option D.



7.如图所示, A、B 两容器容积相等, 用粗细均匀的细玻璃管连接, 两容器内装有不同气体, 细管中央有一段水银柱, 在两边气体作用下保持平衡时, A 中气体的温度为 0°C , B 中气体温度为 20°C , 如果将它们的温度都降低 10°C , 则水银柱将(As shown in the figure, containers A and B have the same volume and are connected by a thin glass tube with uniform thickness. The two containers are filled with different gases, and there is a mercury column in the center of the thin glass tube. When the mercury column is in equilibrium under the action of the gases on both sides, the temperature of the gas in A is 0°C and that in B is 20°C . What will happen to the mercury column if both temperatures are reduced by 10°C ?)



- A. 向 A 移动 (Move to A)
- B. 向 B 移动 (Move to B)
- C. 不动 (Remain stationary)
- D. 不能确定 (Cannot be determined)

答案(Answer):A

答案解析:

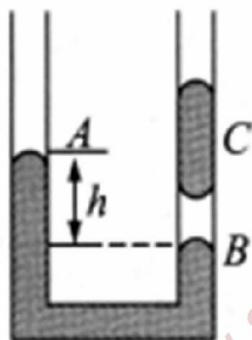
中文:初始水银柱平衡, 故 $p_A = p_B$, 容器容积不变, 两气体均做等容变化, 由查理定律得压强变化量 $\Delta p = \frac{p}{T} \Delta T$ 。换算热力学温度: $T_A = 0 + 273 = 273 \text{ K}$, $T_B = 20 + 273 = 293 \text{ K}$; 温度降低 $\Delta T = -10 \text{ K}$, 则压强减小量 $|\Delta p_A| = \frac{p}{273} \times 10$, $|\Delta p_B| = \frac{p}{293} \times 10$,

A 中气体压强减小量更大。降温后 $p_A' < p_B'$, 水银柱在压强差作用下向 A 端移动。

English: The mercury column is in equilibrium in the initial state, so $p_A = p_B$. The volume of the container is constant, and both gases undergo isochoric change. The pressure change is derived from Charles's Law as $\Delta p = \frac{p}{T} \Delta T$. Convert Celsius temperature to thermodynamic temperature: $T_A = 0 + 273 = 273 \text{ K}$, $T_B = 20 + 273 = 293 \text{ K}$; the temperature decreases by $\Delta T = -10 \text{ K}$, then the pressure reduction $|\Delta p_A| = \frac{p}{273} \times 10$, $|\Delta p_B| = \frac{p}{293} \times 10$, the pressure reduction of gas in A is greater. After cooling, $p_A' < p_B'$, and the mercury column moves to end A under the action of the pressure difference.



8.如图所示，两端开口的 U 形管，右侧直管中有一部分空气被一段水银柱与外界隔开。若在左管中再注入一些水银，则平衡后(As shown in the figure, a U-tube with open ends has a part of air in the right straight tube separated from the outside by a mercury column. What will happen after equilibrium if some more mercury is injected into the left tube?)



- A. h 减小 (h decreases)
- B. h 增大 (h increases)
- C. C. 右侧封闭气柱体积变小 (The volume of the confined air column on the right becomes smaller)
- D. h 不变 (h remains unchanged)

答案(Answer):D

答案解析:

中文:设大气压为 p_0 , 右侧封闭气体的压强 $p = p_0 + \rho g h$ (h 为右侧封闭气柱下方水银柱竖直高度), 且 p 等于 p_0 加上 U 形管左右水银面的高度差。向左管注入水银后, 右侧封闭气体始终与大气压通过水银柱保持压强平衡, **封闭气体压强始终不变**; 环境温度不变, 封闭气体温度也不变, 因此气体体积保持不变, h 始终等于左右水银面高度差, 故 h 不变。

English: Let the atmospheric pressure be p_0 , the pressure of the confined gas on the right is $p = p_0 + \rho g h$ (h is the vertical height of the mercury column below the confined air column on the right), and p is also equal to p_0 plus the height difference of the mercury surface on the left and right sides of the U-tube. After injecting mercury into the left tube, the confined gas on the right is always in pressure equilibrium with the atmospheric pressure through the mercury column, and **the pressure of the confined gas remains constant all the time**; since the ambient temperature is constant, the temperature of the confined gas is also constant, so the gas volume remains unchanged, and h is always equal to the height difference of the left and right mercury surfaces, thus h remains unchanged.